

Experiment 8: Decimation and Interpolation of a Discrete-Time Signal in MATLAB

Aim: To perform decimation by 2 and interpolation by 2 on a discrete-time sinusoidal sequence and compare the original, decimated, and interpolated sequences.

MATLAB Code

```
clc;
clear;
close all;

% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);

% Decimation by 2
y = x(1:2:end);
nd = 0:length(y)-1;

% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;

% Single Figure with All Outputs
figure(1);

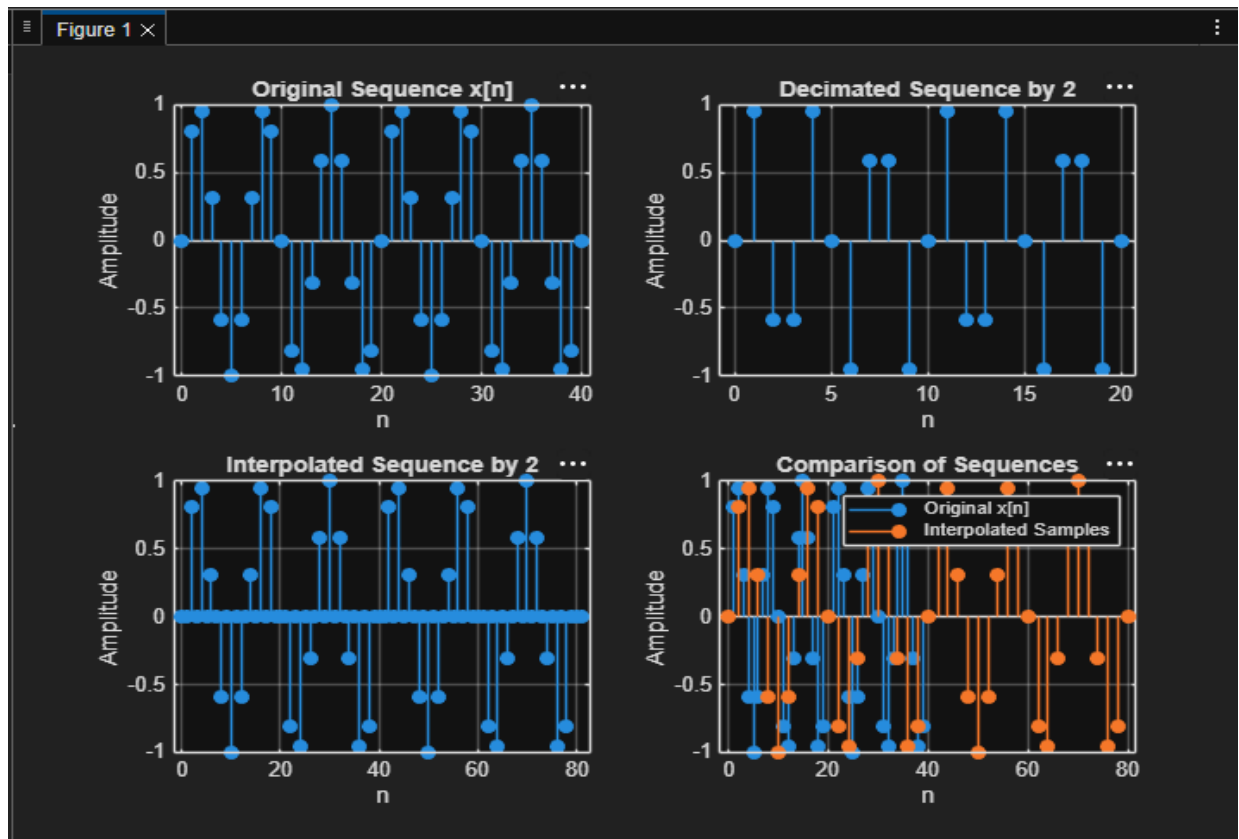
% Original Sequence
subplot(2,2,1);
stem(n,x,'filled');
grid on;
title('Original Sequence x[n]');
xlabel('n');
ylabel('Amplitude');

% Decimated Sequence
subplot(2,2,2);
stem(nd,y,'filled');
grid on;
title('Decimated Sequence by 2');
xlabel('n');
ylabel('Amplitude');

% Interpolated Sequence
subplot(2,2,3);
stem(ni,v,'filled');
grid on;
title('Interpolated Sequence by 2');
xlabel('n');
ylabel('Amplitude');

% Comparison
subplot(2,2,4);
stem(n,x,'filled');
hold on;
stem(0:2:length(v)-2,v(1:2:end),'filled');
legend('Original x[n]', 'Interpolated Samples');
grid on;
title('Comparison of Sequences');
xlabel('n');
ylabel('Amplitude');
```

Output



Result: The original sequence was successfully decimated by a factor of 2 and interpolated by a factor of 2. The resulting sequences were plotted and compared successfully in MATLAB.